

Politecnico di Milano

Prova finale: Introduzione all'analisi di missioni spaziali

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INTRODUCTION

In this project we attempted to identify possible transfer orbits of a spacecraft rotating around the Earth, from a specific initial position (belonging to the initial orbit) to an exact given final position (in the final orbit). The goal is to characterize these orbits in an engineering way, evaluating the costs in terms of impulse and time to complete the mission. All this will be fundamental to decide which of the maneuvers described is most convenient.

The group worked in harmony and in a strategic way, to best express the abilities of each individual member, periodically rechecking the work done by others to verify every procedure. Each member of the group learned how to use GitHub, allowing us to have a better collaboration and a cleaner workflow. The calculator used was MATLAB®, and it was fundamental for implementing functions that allowed the parameters of the transfer orbits to be functionally obtained, subsequently graphing them in 3-D to better understand the actual trajectory performed by the spacecraft. Finally, a scatterplot will allow you to highlight the strengths and weaknesses of each maneuver.

INITIAL AND FINAL ORBIT CHARACTERISATION

Initial orbit characterization

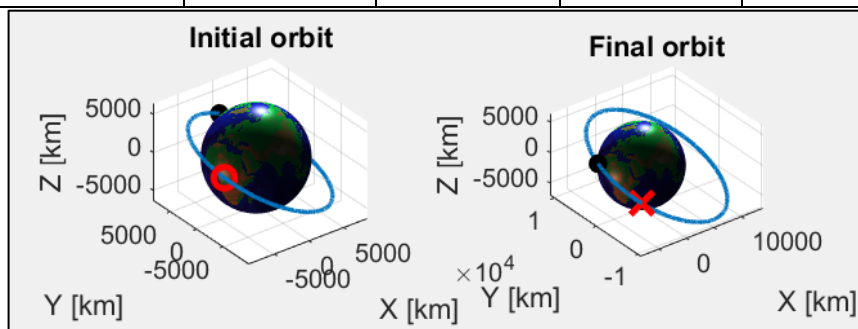
GIVEN DATA	$x_i [km]$	$y_i [km]$	$z_i [km]$	$v_{xi} \left[\frac{km}{s} \right]$	$v_{yi} \left[\frac{km}{s} \right]$	$v_{zi} \left[\frac{km}{s} \right]$
value	-8205.9652	-4682.7991	2382.4314	1.8330	-5.5360	-2.2630

ORBITAL ELEMENTS	$a_i [km]$	$e_i [-]$	$i_i [deg]$	$\Omega_i [deg]$	$\omega_i [deg]$	$\theta_i [deg]$
value	9337.0000	0.1000	27.1545	59.1581	26.7543	120.8517

Final orbit characterization

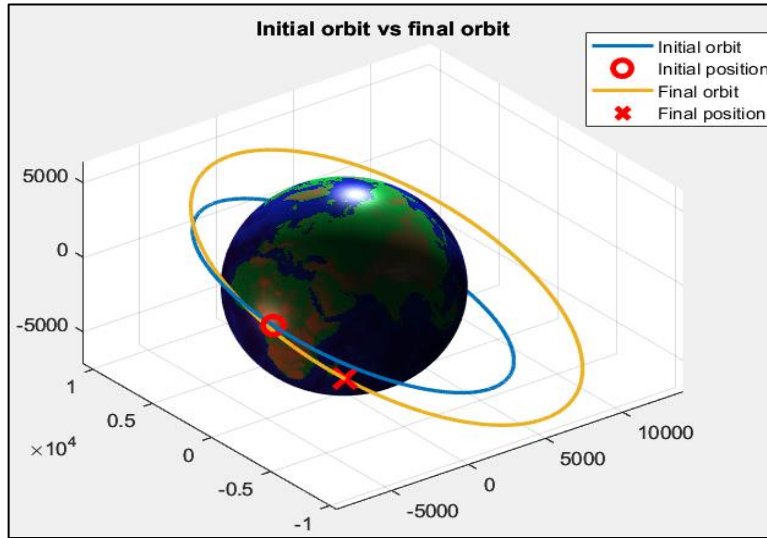
GIVEN DATA	$a_f [km]$	$e_f [-]$	$i_f [deg]$	$\Omega_f [deg]$	$\omega_f [deg]$	$\theta_f [deg]$
value	11860.0000	0.2614	30.6647	45.8653	132.4678	55.6858

ORBITAL ELEMENTS	$x_f [km]$	$y_f [km]$	$z_f [km]$	$v_{xf} \left[\frac{km}{s} \right]$	$v_{yf} \left[\frac{km}{s} \right]$	$v_{zf} \left[\frac{km}{s} \right]$
value	-5795.155	-7660.0788	-696.6091	4.1115	-4.4159	-3.5729



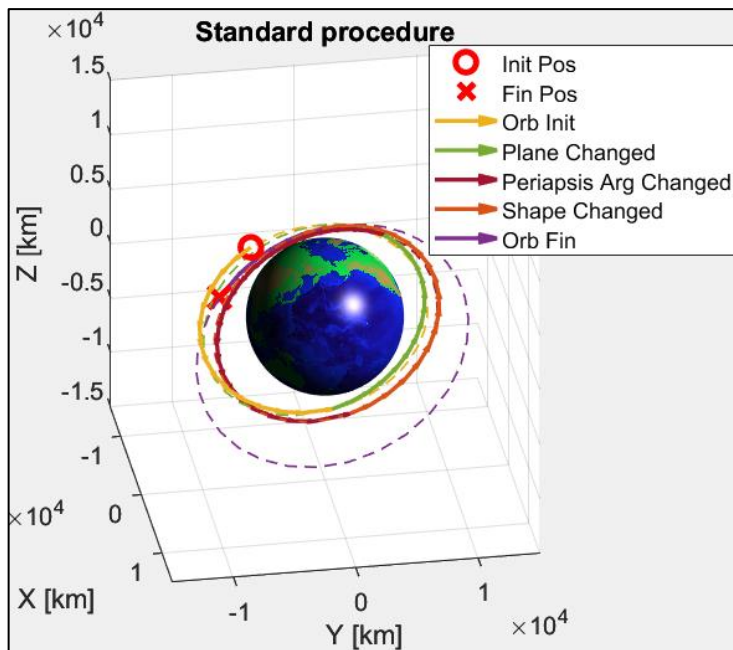
TRANSFER TRAJECTORY DEFINITION AND ANALYSIS

Initial vs final orbit discussion



The initial and final orbits differ in all the orbital parameters, in fact there are no intersections between the two. It is therefore necessary to vary the parameters through standard maneuvers or by implementing a direct maneuver from the starting point to the final point using just two pulses. Having to change the anomaly of the pericenter, shape and plane we will have 6 possible combinations. If we consider that we can change shape either at the pericenter or at the apocenter we will therefore obtain 12 different maneuvers.

Standard strategy



This strategy uses 4 distinct impulses to change orbit from the given initial and final position: the first impulse is designated for the plane change, the second for the secant burn to align with the apsidal axis, and the remaining two involve a bitangent transfer between apoapsis and periapsis. As regards the change of plane, it will be necessary to change both the inclination and the axis of the nodes of the orbit. Therefore, the maneuver does not take place in correspondence with the nodal axis so a rotation will be generated around an axis not lying on the equatorial plane.

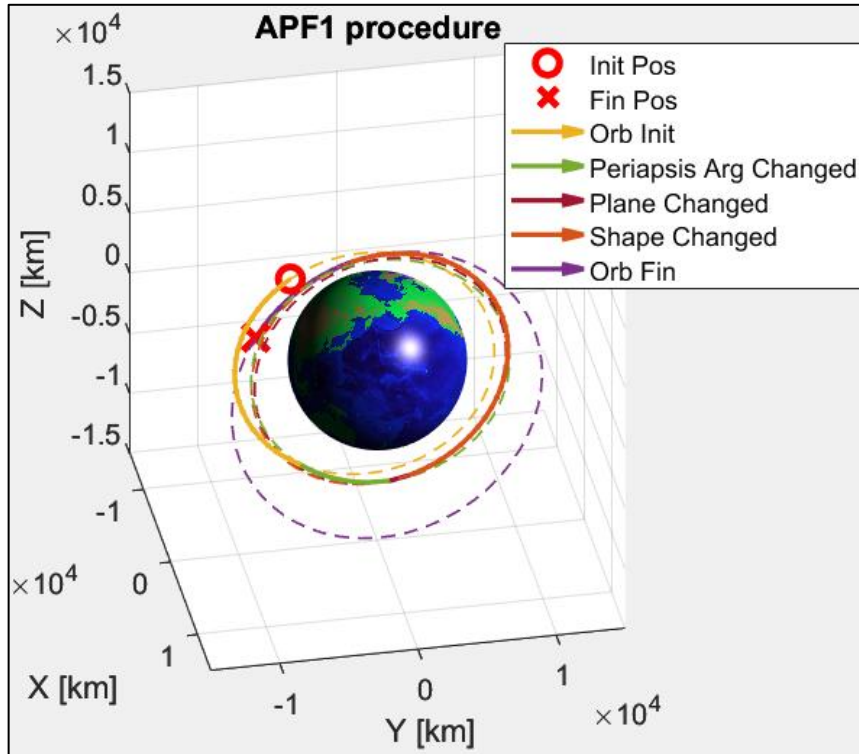
The point where the plane must be changed is defined entirely by geometry, more specifically by the theorem of sines and cosines. There are in general two possible plane-change points, which differ by a value of 180° . The most convenient for

this specific case is the one at $\theta = 265.9517^\circ$ because it has a lower transverse velocity and so a lower Δv is needed to change plane.

The second step is a change in periapsis argument, which is operated at $\theta = 265.9517^\circ$ because of less time spent on the orbit. The last two impulses are given by the bitangent orbit. The total cost of the maneuver, in terms of Δv , is the sum of each impulse Δv . The total time necessary to operate the standard maneuver is the sum of the time spent on every orbit, because each impulse is approximated as instantaneous.

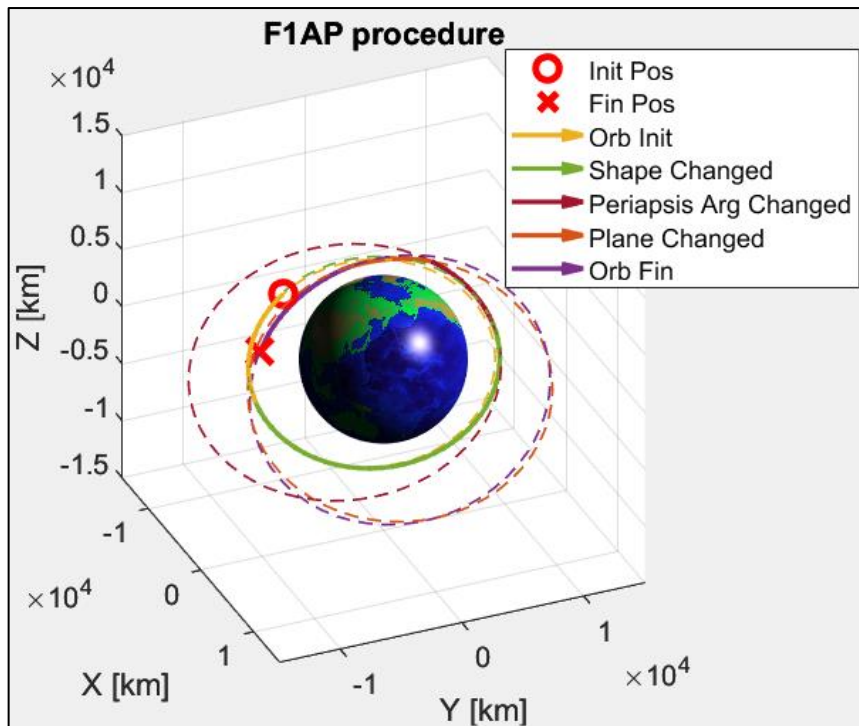
Alternative strategies

1. APF1



terms of total impulse and relatively fast, it is the most efficient among the standard maneuvers with change of initial anomaly.

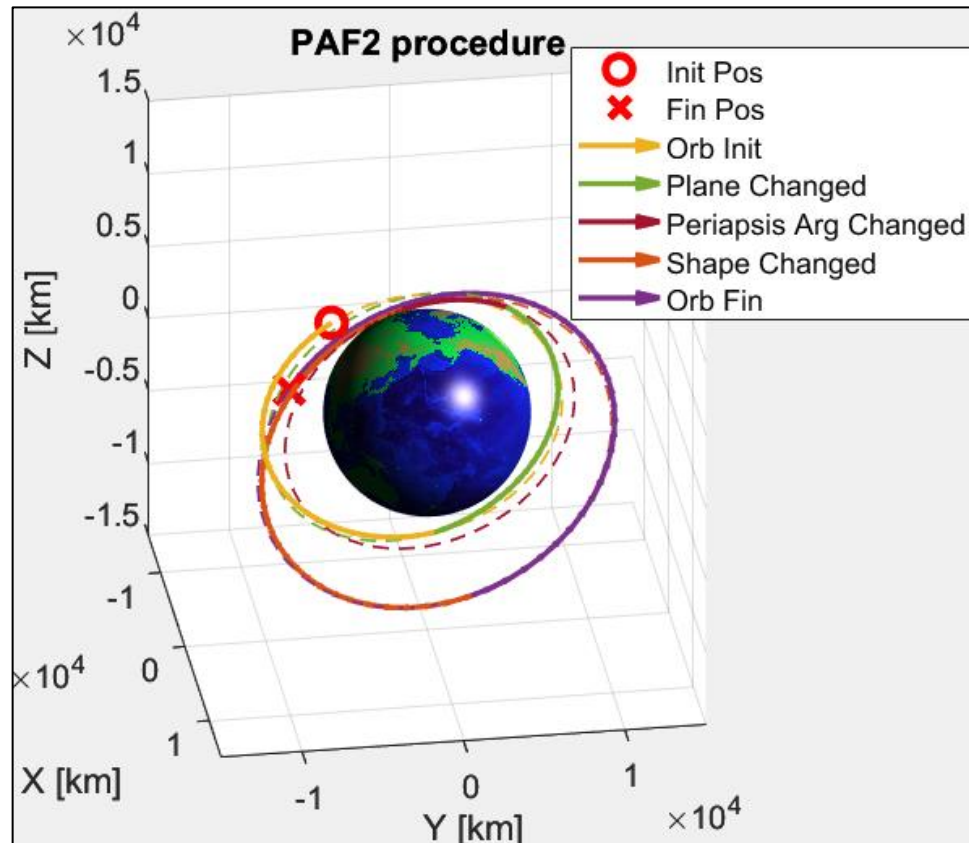
2. F1AP



This is the first of the variations of the standard maneuver that will be illustrated. It is developed by initially changing the pericenter anomaly, in the first available point, by an anomaly different from the final value, since in the next step the change of plane involves a further change in the direction of the eccentricity vector. The first change of anomaly is therefore studied so that at the change of inclination and nodal axis, at the geometrically established point, the value of omega is that of the final orbit. Intermediate turns out to be 120.8139° , which subsequently passes to 132.4678° due to change of plane. The shape change occurs at the apocenter with a bitangent transfer's orbit. The last stretch goes from the pericenter of the final orbit to the arrival point. This maneuver is overall cheap in

This manoeuvre begins with a change of shape at the apocenter, which is more convenient than performing it at the pericentre as it is closer to the release point, and changing shape at the apocenter is cheaper. Also, in this case the change of anomaly of the pericentre occurs at a point defined by the geometry of the orbits through a secant manoeuvre, and it will have to consider the subsequent change of plane, which, as we have already said previously, causes a further change of anomaly. This manoeuvre is shorter than APF1 but requires a larger Δv of just over $1 \frac{km}{s}$.

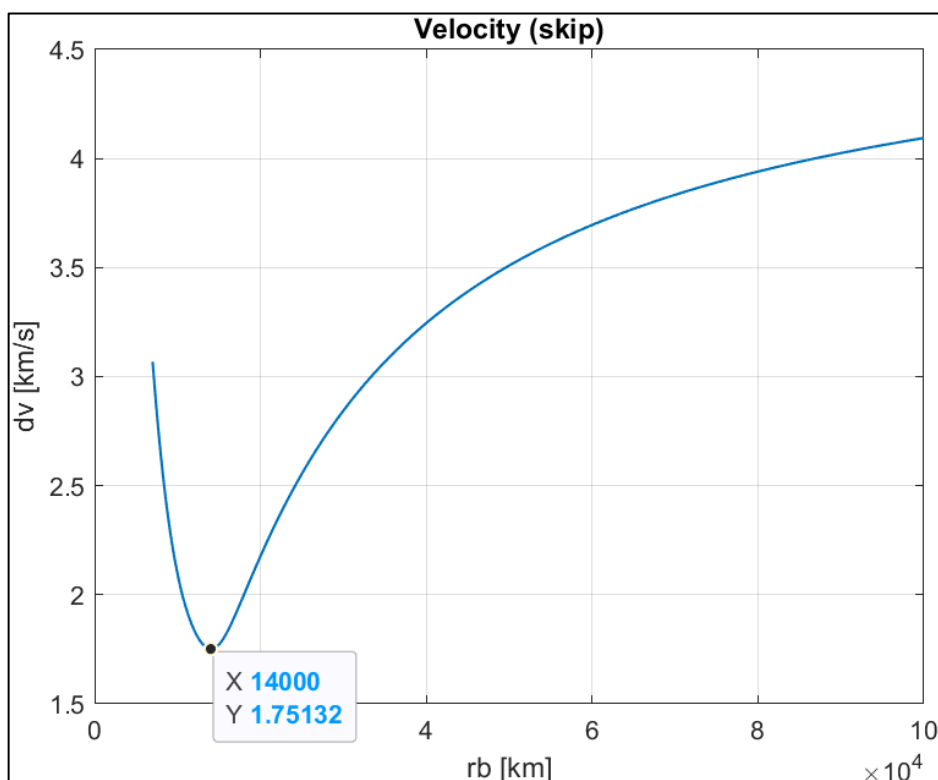
3. PAF2



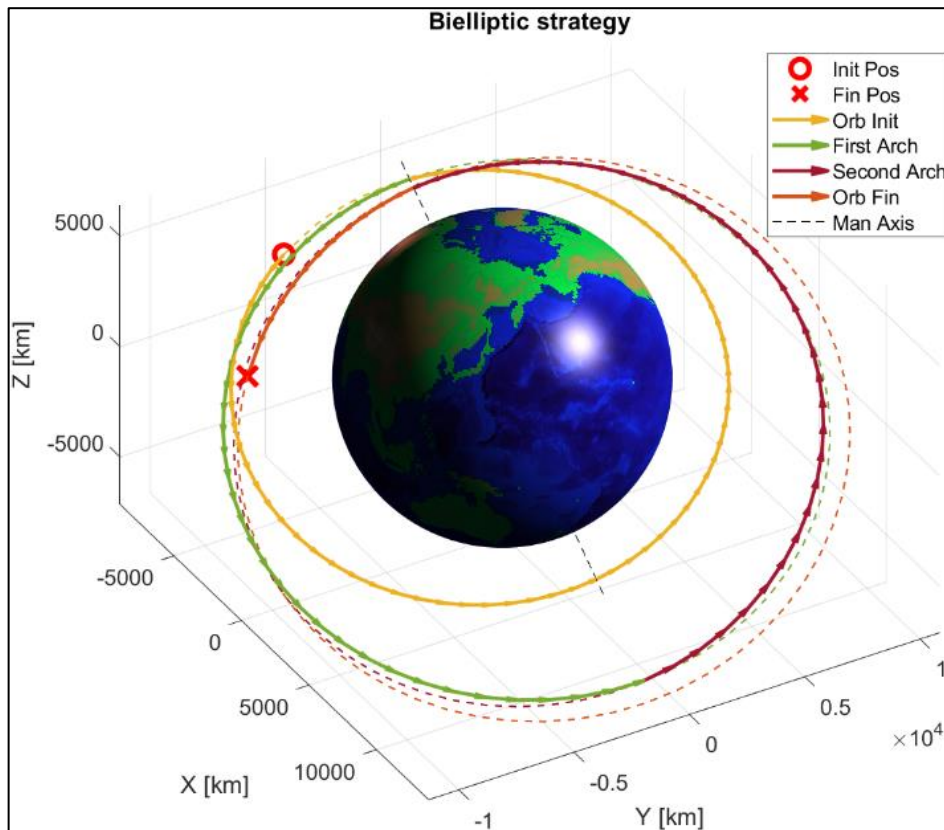
This manoeuvre is identical to the standard manoeuvre but differs only in the way the final shape changes. In fact, if in the standard procedure the bitangent transfer orbit started from the apocenter, in this case it starts from the pericentre to arrive at the apocenter of the final orbit. The first two manoeuvres are obviously a plane change, to vary the inclination and nodal axis, and a secant manoeuvre to vary the pericentre anomaly. This strategy results in a slight decrease in total impulse but requires a longer transfer time of approximately 1 hour. Of the two, the standard manoeuvre illustrated above is therefore more convenient.

Bielliptic transfer

In the formal analysis conducted, the feasibility and cost-effectiveness of bi-elliptic maneuvers were examined based on the required maneuver radius and associated impulse requirements.



The results indicated that bypassing the initial available point for the maneuver is the most economically efficient strategy. Not only does this method slightly decrease the transit time, but it also simultaneously leads to noticeable fuel savings, optimizing overall resource utilization.

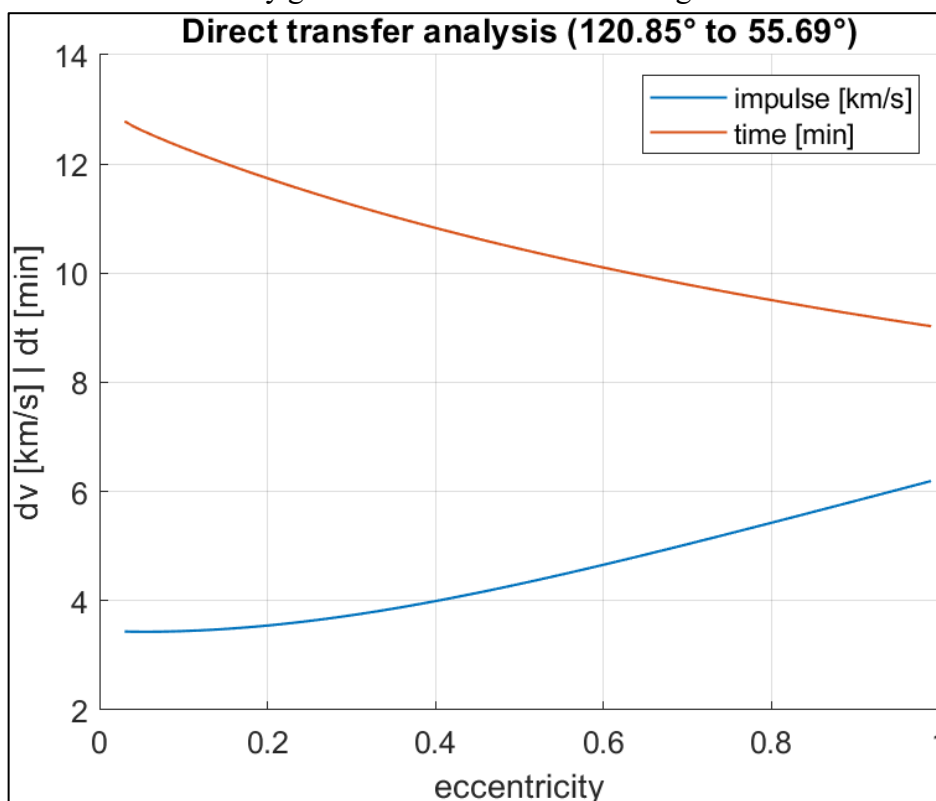


Most economical situation at the first available bi-elliptic maneuver point: $rb = 9300 \text{ km}$ | $\Delta v = 2.02 \text{ km/s}$ | $\Delta t = 7 \text{ h } 20 \text{ m } 57 \text{ s}$

Most economical situation at the second available bi-elliptic maneuver point: $rb = 14000 \text{ km}$ | $\Delta v = 1.75 \text{ km/s}$ | $\Delta t = 6 \text{ h } 59 \text{ m } 59 \text{ s}$

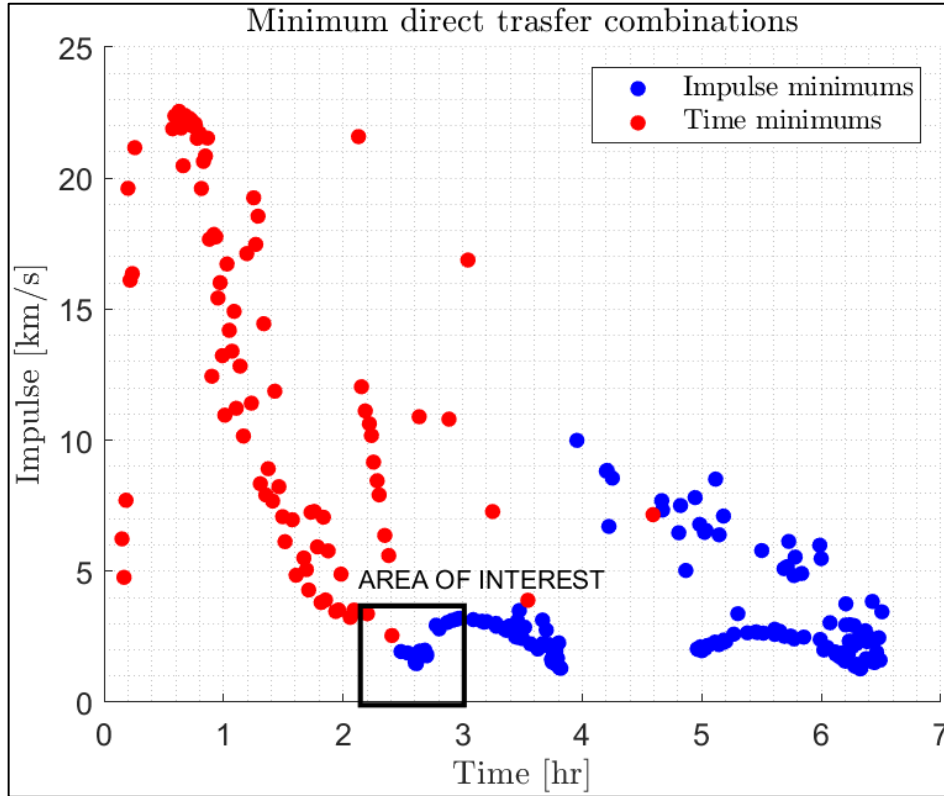
Direct transfer

Direct orbital transfer may be a pivotal time-saving strategy, enabling spacecraft to transition from one orbit to another without intermediate maneuvers, but can also be optimized for being cost-effective. This strategy minimizes fuel consumption and optimizes mission efficiency by executing a direct trajectory between any initial and final orbit for any given anomalies. The following discussion elaborates on the direct orbital transfer strategy,

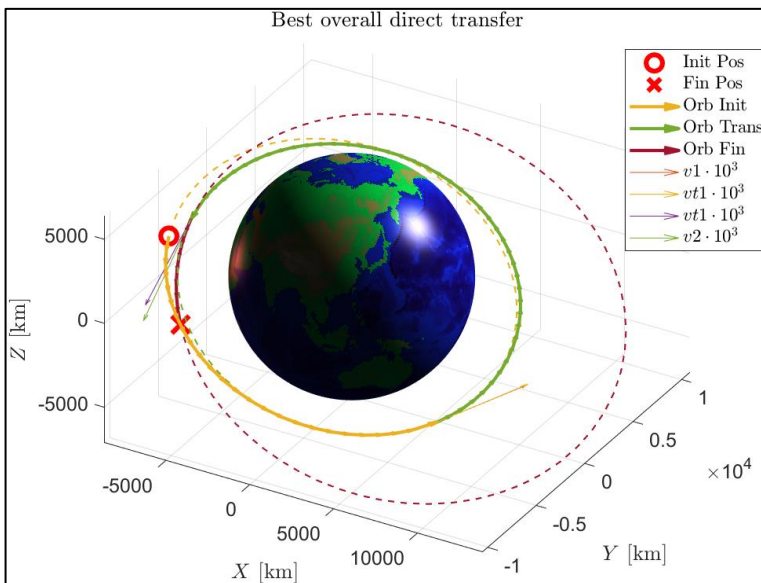


referencing the MATLAB functions **directOrb** and **directOrbLin** for computational insights. By leveraging Keplerian orbital elements, impulse (Δv), flight time (Δt), and other critical parameters required for transitioning between two specified points in space are computed. Transfer orbit's parameters are worked out by either analytically finding value for true maneuvering anomaly in transfer orbit – in case that is the parameter we want to have control over – or numerically solving a single nonlinear equation – in case transfer orbit's eccentricity is chosen as an input parameter for ease of subsequent analysis –. In scenarios where multiple solutions exist - two

distinct complex solutions are typically found in the non-linear case (see graphical solution) - or potential Earth collisions are detected, control parameters for resolving ambiguities and ensuring mission safety and maximum efficiency are provided, and appropriate algorithms are implemented. Multiple manoeuvring options such as perigee-wise, apogee-wise, or best solution are offered, in addition to warnings issuance. A specific algorithm is implemented in order to select correct sense of Results for iterative trials on direct transfer from given initial true anomaly to given final true anomaly were obtained here as an example (see figure).



Initially, parameters such as angle and eccentricity resolutions are set up, along with defining boundaries for these variables. Subsequently, an optimization task is executed using nested loop structures to explore various combinations of these parameters. Optimization functions, specifically **fminsearchbnd**, are employed within these loops to determine the minimum velocity and time combinations required for the transfer, considering set constraints and objectives. After optimization, the results are organized and ranked based on assigned importance to velocity and time. A scatter plot is generated as a visual representation to compare optimal solutions concerning impulse and time (cfr. figure).



Ultimately, the best overall direct transfer is identified by prioritizing criteria related to velocity and time, yielding specific angles and eccentricities for the optimal maneuver.

Best overall direct transfer: from 244.75° to 15.43° at $e = 0.0934$

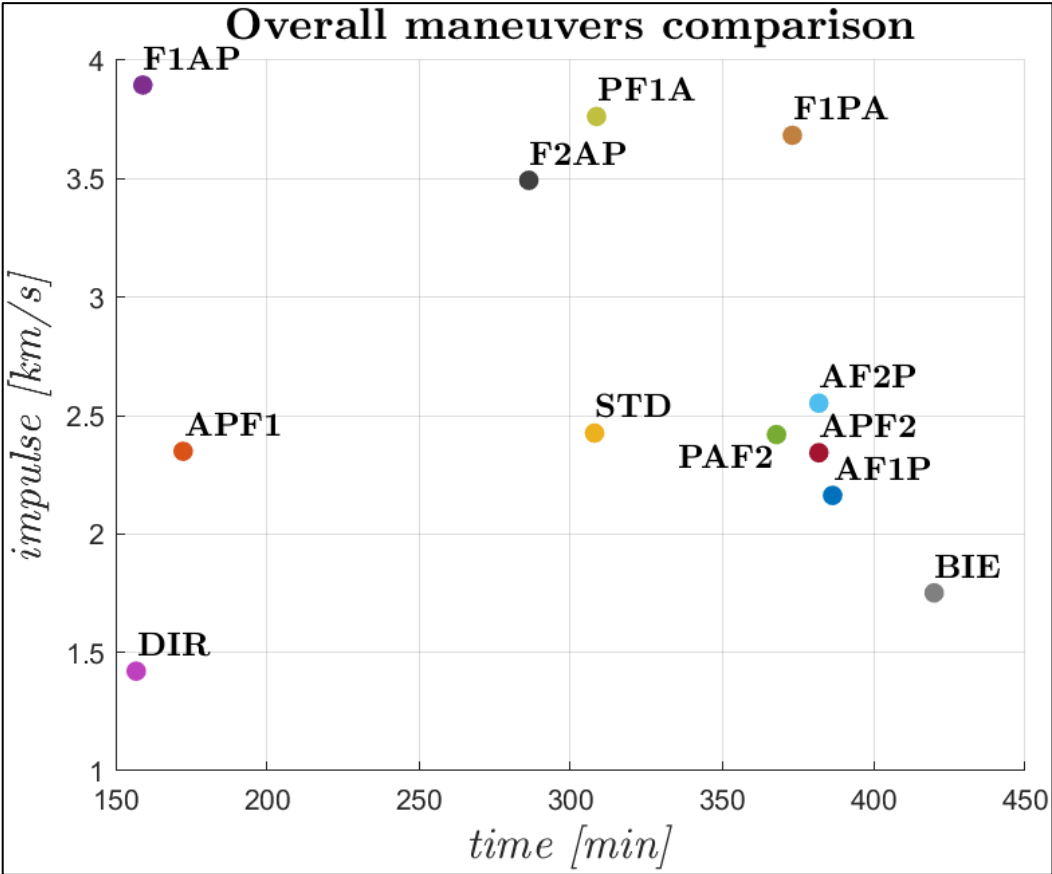
Δv : 1.42 km/s

Δt : 9407 s = 2 hr 36 min 47 s

(a resolution of $10^\circ \times 0.1$ produced 12,321 cost-oriented maneuvers and an equivalent number of time-oriented ones, yielding a grand total of 24,624 possible raw combinations, among which such result was selected)

DISCUSSIONS AND CONCLUSIONS

The evaluation of different orbital maneuver strategies has revealed crucial insights for optimizing spacecraft trajectory changes. Fuel efficiency, time considerations, and mission objectives should be analyzed for each maneuver, highlighting their respective strengths and weaknesses. The trade-off between fuel consumption and time efficiency is a key factor in mission planning. The identified optimal maneuver, the direct one, provides a valuable benchmark, guiding planners in the selection of strategies based on specific mission parameters.



The initial orbits were not significantly disparate from one another, suggesting a relatively consistent starting point for analysis. In general, the alternatives to the standard maneuver did not yield substantially more favorable outcomes, with only a few exceptions observed.

However, through iterative optimization processes, notable and more advantageous results were achieved, underscoring the value of such iterative approaches in the study and refinement of orbital maneuvers.

This study contributes practical insights for space mission planning, emphasizing the importance of considering both fuel and time efficiency in orbital maneuvers.

APPENDIX

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